VITAN CONSULTANTS

ARCHITECTS & INTERIOR DESIGNERS

5TH FLOOR, MIDTOWN EMBASSY, BEHIND VASANT VALLEY BUS DEPOT,
OPP. NARAYANA SCHOOL, VASANT VALLEY, GANDHARE KALYAN (W).

TEJAS ANIL NIRGUDE
B. ARCH & INTERIOR DESIGNER

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M.:8451985999

संदर्भ क्र.९६१
दि.११/०३/२०२६

प्रति,
मा.कार्यकारी अभियंता
पाणी पुरवठा विभाग
क.डो.म.पा
कल्याण.

विषय :- स.नं.६७/१/२,स.नं.६७/१/३,स.नं.६८/६,स.नं.६८/७,स.नं.६८/८ मौजे: दावडी , ता. कल्याण, मे.रिंजेन्सी निर्माण लि या भुखंडावर बांधण्यात येणाऱ्या इमारतीच्या बांधकाम परवानगीच्या दाखल्यासाठी पाणी पुरवठा विभागचा नाहरकत दाखला मिळणे बाबत.

१) भूखंडाचे विवरण- स.नं.६७/१/२,स.नं.६७/१/३,स.नं.६८/६,स.नं.६८/७,स.नं.६८/८ मौजे: दावडी,ता.कल्याण,
२) विकसीत करावयाचे क्षेत्र – १९७४०.००चौ.मी.चौ.मी.प्रस्तावित सदनिका ११६० आहेत व गाळे ११ आहेत
३) आवश्यक पाणी पुरवठा ८,००,०००.००लिटर
४) STP मधुन Recycle द्वारे वापरात येणारे पाणी २,५०,०००.००लिटर
५) महापालिकेकडून आवश्यक पाणी पुरवठा ५,५०,०००.००लिटर
६) पाण्याच्या टाकीची लांबी, रुंदी, उंची :- सवीस्तर लेआउट सोबत जोडला आहे.
७) इमारती वरील पाण्याच्या टाकीची लांबी रुंदी, उंची :- सवीस्तर लेआउट सोबत जोडला आहे.
८) पाणी पुरवठा लाईन पासुन U.G Tnak चे अंतर ९०.०मी

वितान कन्सल्टंट करिता

वास्तुशिल्पकार.

WATER SUPPLY SYSTEM DESIGN

FOR PROPOSED BUILDING ON PLOT BEARING ON PROPOSED S.NO:67/1/2, 67/1/3,68/6,68/7,68/8 BUILDING VILLAGE-DAVDI, TALUKA-KALYAN, DIST-THANE FOR REGENCY NIRMAN LTD. ULHASNAGAR

PRESENTED TO: KALYAN - DOMBIVLI MUNICIPAL CORPORATION

PREPARED FOR:

REGENCY NIRMAN LTD

JANUARY 2026

PROJECT: REGENCY NEO, DOMBIVLI

PREPARED BY: ENGINEERING CREATIONS PUBLIC HEALTH CONSULTANCY

Address: Office No. 703, 8th Floor Ecstacy Business Park, Near Mulund East-West Flyover, Jatashankar Dosa Road, Mulund (West), Mumbai, Maharashtra 400080

Email id: engcreat@ecphc.com

WATER SUPPLY SYSTEM

1.1 PROPOSED STORAGE: (UNDER GROUND)

- The source of water supply will be municipal water supply which will be used for domestic purpose. The water from the municipal water supply line will be directly fed into the proposed underground fire tank through appropriate ball and float valve. The municipal water from the fire tank will overflow into the domestic tank.
- The baffle wall arrangement inside the fire tank will keep the fire tank water in circulation.
- The underground water tank is proposed below ground floor. Separate compartment for flushing water supply will be provided. The recycled treated water from sewage treatment plant will be supplied to the underground flush water tanks.
- The underground water tanks will comprise of the following compartmental storages.
  a. Fire tank.
  b. Domestic storage.
  c. Flushing storage

1.2 DESIGN CONSIDERATIONS:

- The system will be designed to provide pressure of at least 1kg/cm^2 at remotest outlet. All piping will be sized to restrict velocity to less than 2.40m/s and 1.20 m/s for branches supplying quick closing valves.

1.2.1. Domestic / Flushing Water Demand:

- Please refer table no 1, for Water demand and storage water calculations.

  Domestic demand per day: 90 lpcd

  Flushing demand per day: 45 lpcd

  Total water demand per day: 135 lpcd

(As per NBC 2016, VOL 2, PART 9 PLUMBING SERVICES, CLAUSE 4.1.1, WATER REQUIREMENT FOR RESIDENCIES)

1.2.2. Fire fighting tank:

- Fire Reserve Storage

(As per NBC 2016, VOL 1, PART 4 FIRE AND LIFE SAFETY, CLAUSE 4.9, TABLE NO. 7, (A-c) RESIDENTIAL BUILDINGS)

1.3 DISTRIBUTION SYSTEM:

- The water from underground tanks will be pumped into the respective overhead storage tanks provided at terrace level. The underground tanks will be common for all the Buildings.
- Transfer pumps will be separate for domestic and flushing.

1.3.1 Internal distribution

- All concealed pipes will be cPVC Pipes. All exposed pipes in ducts will be uPVC plastic pipes. Proper isolation valves, air vent valves and drain valves will be provided in system.

1.4 WATER DEMAND AND STORAGE CALCULATION

Sr	Description	Comm. BUA Area/flats nos	Occupancy load	Total occupancy	Domestic Water Demand	Flushing Water Demand	Expected sewage Generation	Total Water Demand / flat	UG tank in litres (100%)			O/H tank in litres (50%)		
					90	45	90%		Domestic	Flushing	**Fire	Domestic	Flushing	**Fire
	ZONE 1													
1	BUILDING 1 (G+26)													
	1 BHK	208	4	832	74880	37440	101088	800	1,11,488	54,912		55,744	27,456	25,000
									Provide 115000 ltr	Provide 60000 ltr	3,00,000	Provide 60000 ltr	Provide 30000 ltr	
2	BUILDING 2 (G+26)													
	1 BHK	208	4	832	74880	37440	101088	800	1,11,488	2,19,648		55,744	1,09,824	25,000
									Provide 115000 ltr	Provide 60000 ltr		Provide 190000 ltr	Provide 95000 ltr	
3	COMMERCIAL Building 2													
	ground floor	192.69	3	64	1606	1285	2601	45	1,606	1,285		803	642	
	1st floor	188.95	6	31	787	630	1275	45	787	630		394	315	
	Visitors	15%		14	72	144	194	15	72	144		36	72	
				110	2465	2058	4071		2,465	2,058		1,232	1,029	
									Provide 5000 ltr	Provide 5000 ltr		Provide 3000 ltr	Provide 3000 ltr	
A	TOTAL	416		1774	152225	76938	206247							
							STP capacity 220KLD with area of 176sq.m.							
	ZONE 2													
4	BUILDING 3 (G+22)													
	2 BHK	176	5	880	79200	39600	106920	800	94,336	46,464		47,168	23,232	25,000
									Provide 100000 ltr	Provide 50000 ltr	3,00,000	Provide 240000 ltr	Provide 120000 ltr	
5	BUILDING 4 (G+21)													
	1 BHK	168	4	672	60480	30240	81648	800	90,048	44,352		45,024	22,176	25,000
									Provide 100000 ltr	Provide 50000 ltr		Provide 190000 ltr	Provide 95000 ltr	
6	BUILDING 5 (G+22)													
	2 BHK	176	5	880	79200	39600	106920	800	94,336	46,464		47,168	23,232	25,000
									Provide 100000 ltr	Provide 50000 ltr		Provide 240000 ltr	Provide 120000 ltr	
7	BUILDING 6 (G+28)										3,00,000			
	1 BHK	224	4	896	80640	40320	108864	800	1,20,064	59,136		60,032	29,568	25,000
									Provide 125000 ltr	Provide 65000 ltr		Provide 190000 ltr	Provide 95000 ltr	
8	CLUBHOUSE													
	G+2	1200	1.4	857	21429	17143	34714	45	21,429	17,143		10,714	8,571	
									Provide 25000 ltr	Provide 20000 ltr		Provide 150000 ltr	Provide 10000 ltr	
B	TOTAL	744		4185	320949	166903	439066							
							STP capacity 450 KLD with area of 360sq.m.							
	NET TOTAL (A+B)	1160		5959	473173	243841	645313							
9	COMMERCIAL Building 1													
	ground floor	1302.15	3	434	10851	8681	17579	45	10,851	8,681		5,426	4,341	10,000
	1st to 3rd floor	4071.84	6	679	16966	13573	27485	45	16,966	13,573		8,483	6,786	
	Visitors	15%		167	835	1669	2253	15	835	1,669	1,50,000	417	835	
C	TOTAL			1280	28652	23923	47317		28,652	23,923		14,326	11,961	
							STP capacity 50KLD with area of 40sq.m.		Provide 40000 ltr	Provide 35000 ltr		Provide 20000 ltr	Provide 17000 ltr	

** Fire Fighting Storage shall be fulfilled as per CFO NOC.
Water supply demand for commercial considered is 45 litres/person/day
Water supply demand for residential considered is 135 litres/person/day

With regards,

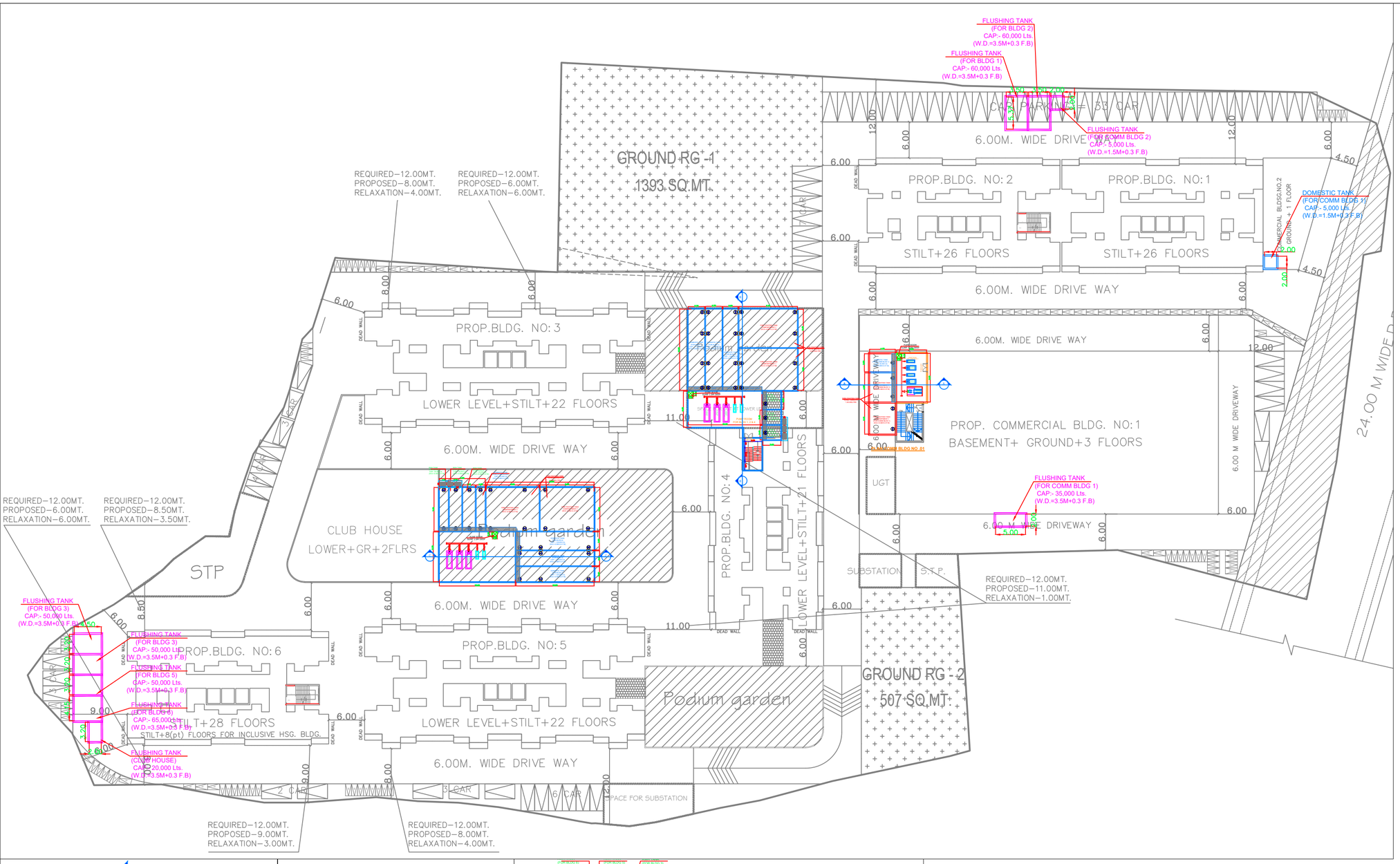
For E.C.P.H.C.P.L.

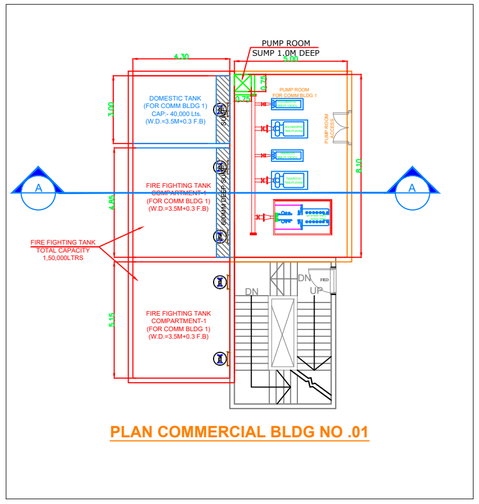
MEP Consultant

Digitally signed by K.C.Jadhav
Date: 2026.02.06 16:09:43 +05'30'

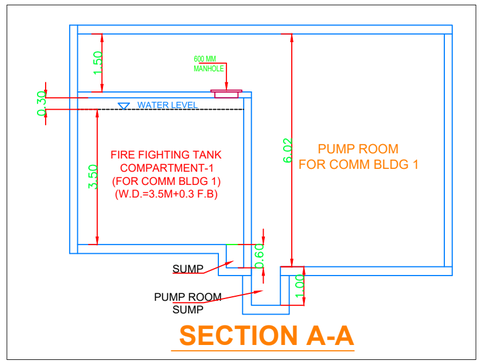
E.C.P.H.C.P.L. MULUND

Consultant Signature

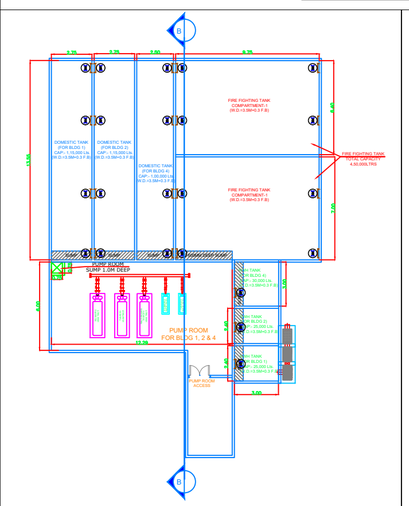




PLAN COMMERCIAL BLDG NO .01

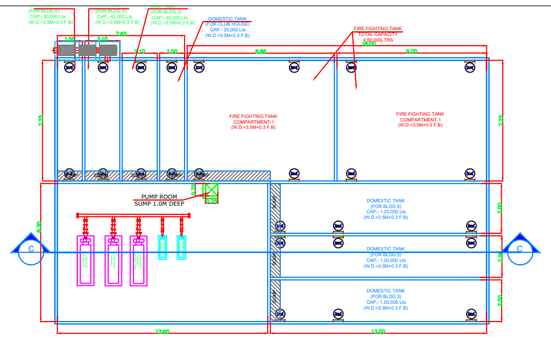


SECTION A-A

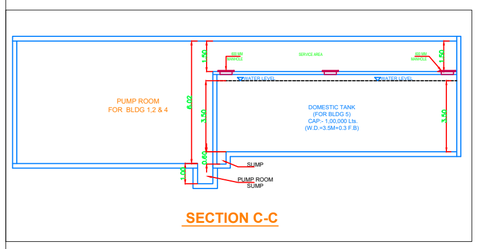


PLAN BLDG NO .01,02 & 04

SECTION B-B



PLAN BLDG NO .03,05, 06 & CLUB HOUSE



SECTION C-C

ARCHITECT–

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PLUMBING CONSULTANT:–

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Mulund (West), Mumbai–400080
Tel. 022 25918279/25918227 TELE–FAX. 25918203
E–mail :engcreat@vsnl.com

PROJECT:–

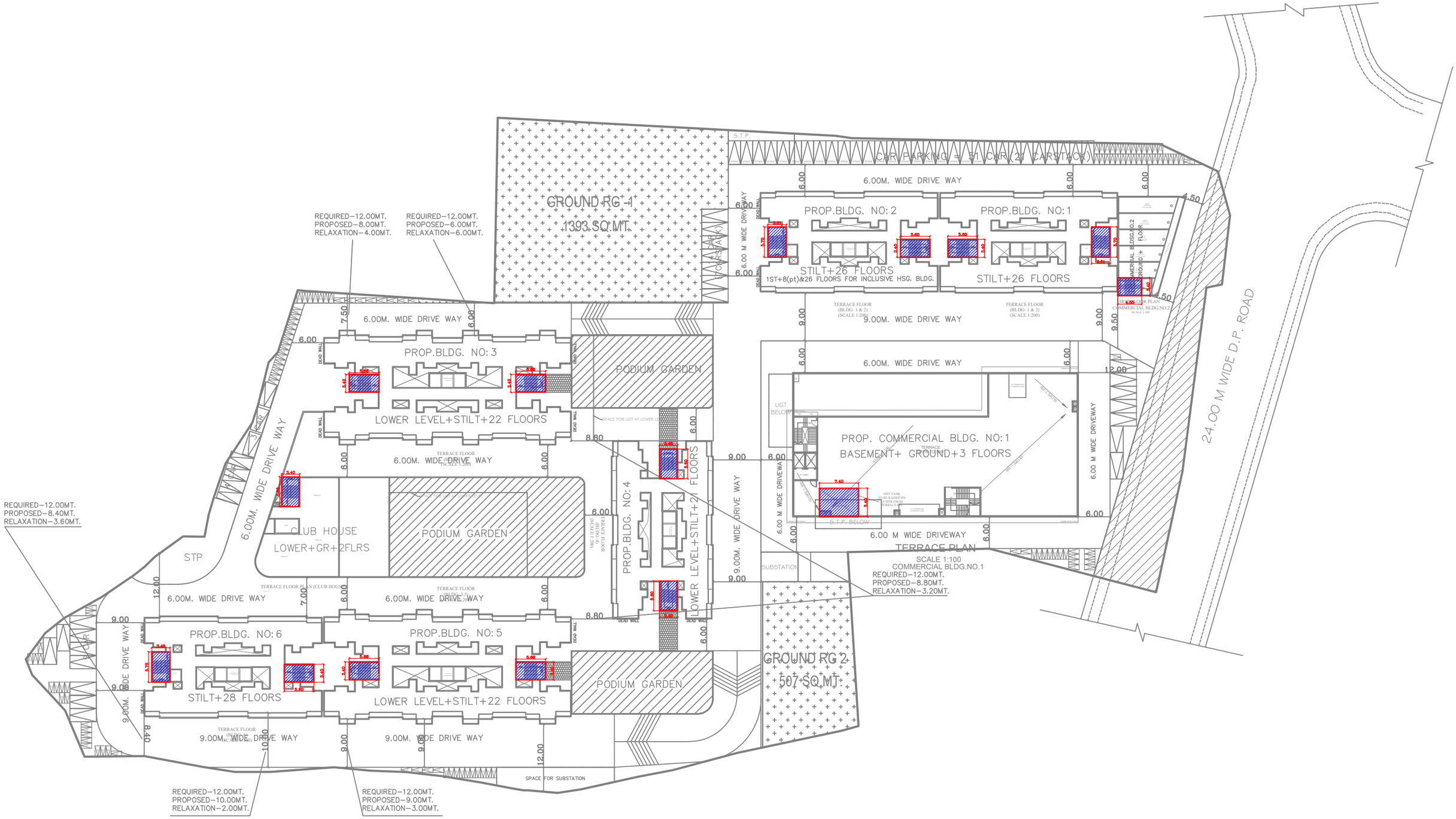
PROPOSED BUILDING LAYOUT ON PLOT BEARING ON S.NO:67/1/2, 67/1/3,68/6,68/7,68/8 VILLAGE–DAVDI, TALUKA–KALYAN, DIST–THANE FOR

TITLE:–

UNDERGROUND WATER TANK LOCATION LAYOUT

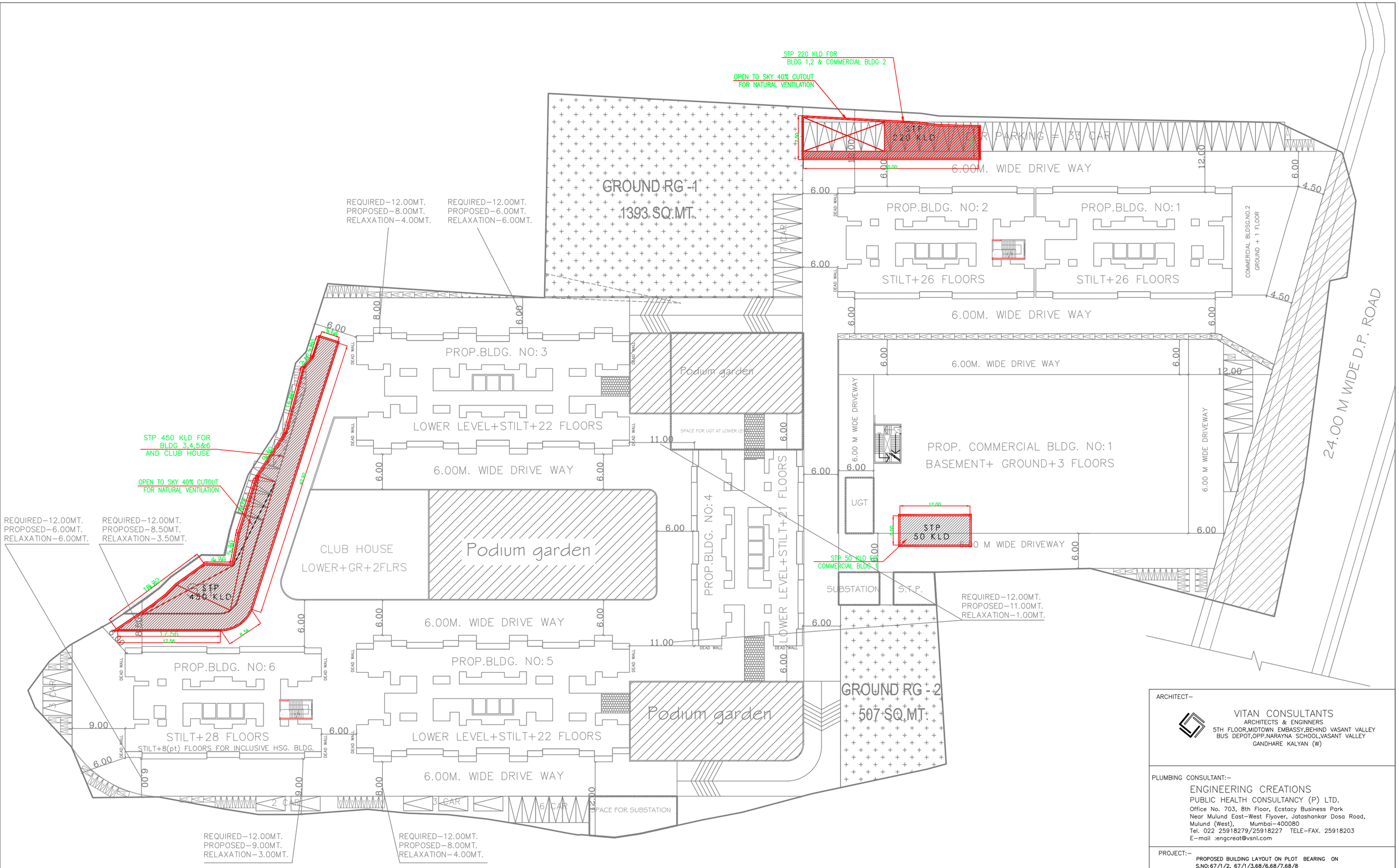
DATE	SCALE	REV. NO.	DRAWN BY	CHKD. BY
09.02.2026	NTS	R.0	GANESH	KAVI

DRG. No.	CAD. No.
REGENCY GROUP DEMART/ECPH/ UNDERGROUND TANK LOCATION LAYOUT	REGENCY GROUP DEMART/ECPH/ UNDERGROUND TANK LOCATION LAYOUT



OHT LAYOUT PLAN

OVERHEAD TANK



LAYOUT PLAN

SCALE- 1:500

ARCHITECT-

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PROJECT:-

PROPOSED BUILDING LAYOUT ON PLOT BEARING ON S.NO:67/1/2, 67/1/3,68/6,68/7,68/8 VILLAGE-DAVDI, TALUKA-KALYAN, DIST-THANE FOR

TITLE:-

STP LOCATION LAYOUT

DATE	SCALE	REV. NO.	DRAWN BY	CHKD. BY
09.02.2026	NTS	R.0	GANESH	KAVI

DRG. No.
REGENCY GROUP DEMART/ECPH/ STP LOCATION LAYOUT

CAD. No.
REGENCY GROUP DEMART/ECPH/ STP LOCATION LAYOUT